

Automated structural design optimization of steel reinforcement using graph neural network and exploratory genetic algorithms

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ABSTRACT

Structural design optimization for steel reinforcement (rebar) is a critical part of reinforced concrete structures. In practice, this process is conducted manually or semi-automatically for each element, which is time-consuming and relies on engineers' knowhow and experience for buildability improvement, material savings and clash resolution. This paper presents an automated pipeline for clash-free rebar design optimization integrating graph neural networks (GNN) and exploratory genetic algorithms (EGA). Graph representation is adopted to characterize rebar layouts in both structural elements and different kinds of reinforced concrete joints. GNN leverages graph representation of rebars to consider the parametric relationship between different rebar groups in a single structural element or among multiple structural elements, enhancing the efficiency of clash-free rebar design optimization. EGA supports design checking and further optimization according to building codes to achieve the optimal design. Compared with conventional optimization methods, the proposed pipeline can automatically identify optimal clash-free rebar design in compliance with code-stipulated requirements while reducing 75%–90% of the computation time, which shows enormous potential for practical use in the industry.

1. Introduction

Reinforced concrete (RC) structure is the most common structural form. The cost of RC structures is composed of the costs of steel reinforcement, concrete and formwork [1]. As concrete and formworks remain almost unchanged after the size of the structural component is determined, structural design optimization for steel reinforcement, which involves specifying the positions and diameters of steel reinforcement (which is also called rebar) in RC elements, is critical to improvement of buildability and cost efficiency [2]. The design of steel reinforcement for RC structures relies on the knowledge and experience of engineers. This process is performed manually or semi-automatically with computer-aided design software, which is time-demanding and sometimes error-prone. Studies have also shown that optimization of the rebar design can significantly reduce the material cost, construction time, and waste generation [3–5]. Though previous work has introduced Design for Manufacturing and Assembly (DfMA) [6] and Building Information Modeling (BIM) [7,8] into steel reinforcement design and showcased that practical designs can be achieved [9], design optimization based on metaheuristic algorithms such as genetic algorithms (GAs) [10,11] is time-consuming and cannot progressively learn from

the previous design generation process. This is caused by the mechanism of metaheuristic algorithms, that is, for each optimization process, these algorithms will initialize with randomly generated population, and there is no way for them to learn from the experience of optimizing previous problems in order to accelerate the optimization process of the next problem.

A distinguishing feature of steel reinforcement design is that the steel reinforcements will across several sections, and the layouts of different rebar groups will affect each other. Fig. 1 provides an example of a RC beam, where the number of rebars in one group will affect that of the related rebar group. However, conventional design optimization methods for steel reinforcement at section level usually provide design according to the structural analysis results such as bending moment, axial force and shear force for several typical sections, which fails to consider the parametric relationship of rebar layout across different sections and between multiple structural elements, leading to poor buildability, material waste, and sometimes non-code-compliant design. There is a lack of systematic consideration and clear representation of the interrelationship of rebars across different sections and between multiple structural elements.

Some emerging machine learning techniques provide a turning point

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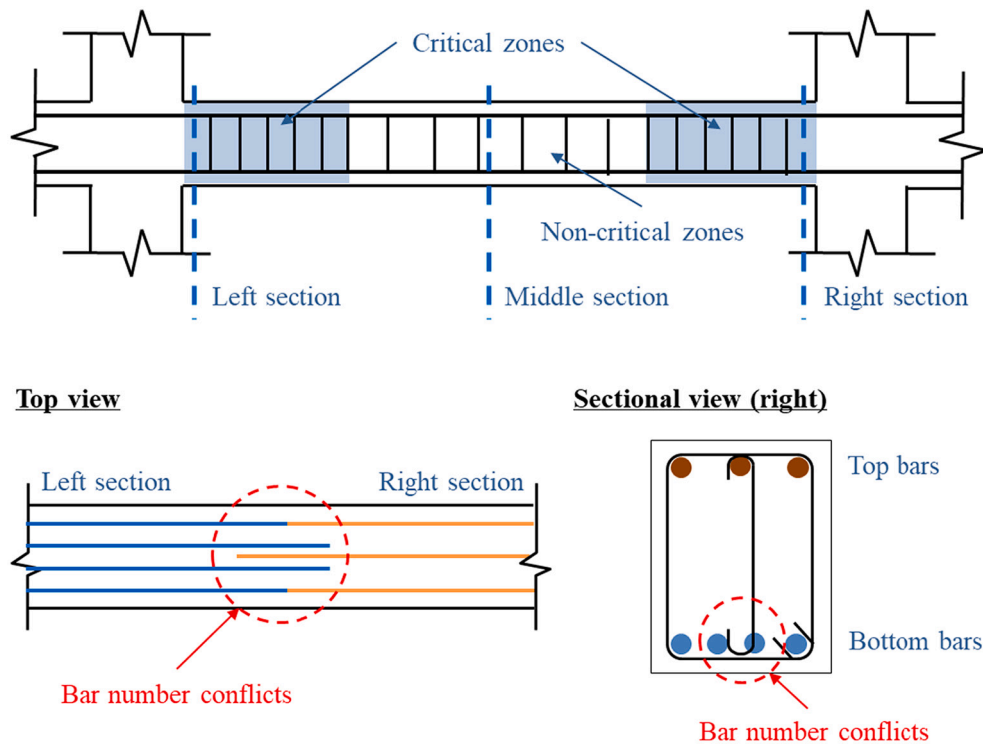


Fig. 1. Example of RC beam: rebar layout of one section will affect other.

for solving the abovementioned issues. Graph neural networks (GNNs) have been proven tremendously successfully for handling unstructured data like email and social website, by providing the graph structure to define the relationship between these data [12,13]. Graphs are helpful in defining concepts that are abstract, like relationships between entities, which coincides with the features of the rebar design problem. In this regard, formulating the rebar design problem in graphs and applying graph neural network to solve it would have great significance. However, since GNN is essentially a form of applied statistics, it cannot guarantee that all the design constraints will be satisfied. Therefore, a feasible solution is to combine GNN with metaheuristic algorithm to leverage strength of both approaches to efficiently explore the search space of rebar design while keeping compliance with all the design requirements.

Therefore, this study aims to develop an automated structural design optimization method integrating GNN. An innovative algorithm with high robustness called exploratory genetic algorithm (EGA) is also developed based on GA to conduct code-compliance checking and further optimization of the design provided by GNN. A design optimization framework incorporating GNN is presented first to demonstrate the conversion of building geometric information and structural analysis into graph representation. Following this, graph-structured data are further utilized to support design optimization and clash diagnosis. Based on the graph representation with element's design information embedded, a rebar design proposal module is developed using GNN to allow rapid rebar design to satisfy the need of industrial practice. Besides, a GNN-based rebar clash diagnosis module is developed to quickly identify and resolve clashes in rebar design. Workflow and implementation details of the key modules are presented with examples. Last but not least, an illustrative example of a RC frame structure is presented to illustrate the proposed design approach, and the results show that the approach can generate practical rebar designs comparable to those generated by metaheuristic algorithms with much less time. The uniqueness of this study includes: (1) the automated design optimization for steel reinforcement using machine learning technique and metaheuristic algorithm, (2) the graph representations of rebar layouts and

their interrelationship in both elements and joints, (3) the first integration of GNN and EGA in rebar design optimization for higher efficiency, (4) the innovative formulation and data-driven model for automated rebar clash diagnosis and resolution. The study lays down the foundation of graph representation of rebar layouts in both elements and joints, facilitating the adoption of graph-based techniques like GNN for rebar design and optimization. With thorough consideration of the industrial requirements on the rebar design efficiency and buildability, the approach proposed in this paper with detailed workflow and implementation steps, can be useful for software development in the future to assist engineers in daily design and reduce rework and save time and cost on the construction site.

The rest of the paper is organised as follows. Section 2 reviews previous relevant literature on steel reinforcement clash avoidance and design optimization using GNN and metaheuristic algorithms. Section 3 presents the methodology details including the problem formulation, GNN and EGA for design optimization. Section 4 presents experimental results and an illustrative example. Section 5 concludes the whole paper and discusses the future work.

2. Literature review

2.1. Steel reinforcement design

Steel reinforcement not only affects the material cost of a concrete structure, but also impacts its labor cost, which can make up over 30% of the total construction cost [14]. Though previous study tended to focus on the material efficiency when carrying out design optimization for steel reinforcement [1,15,16], some researchers start to pay more attention to the buildability of design in recent years [9,17], due to its effect on the structural behavior of RC structure and the productivity of onsite construction. In this regard, improving steel reinforcement design can not only save material cost but also enhance the buildability for greater automation and productivity of construction.

A notable characteristic of rebar detailing design is that the required steel areas of all sections should be satisfied by rebars going through

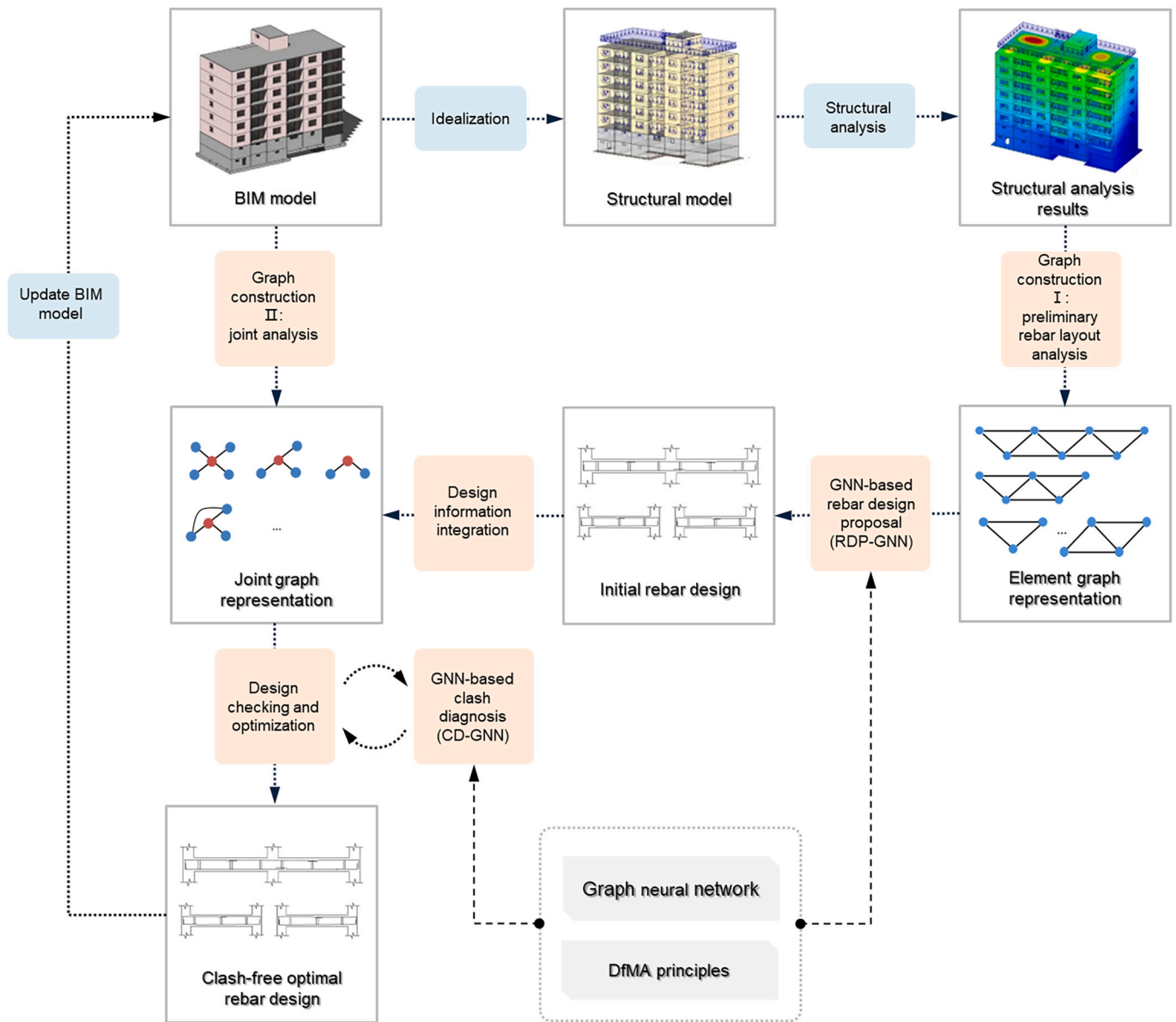


Fig. 2. Proposed framework for clash-free optimal rebar design using GNN.

these sections. In industrial practices, several typical sections are identified for an element for rebar detailing design. For example, for a span of beam, usually three typical sections (right end, middle span and left end) are selected to represent the numerical distribution of steel areas, as shown in Fig. 1, and the rebar layout of one section will affect other. Besides, for each section, the top rebars and bottom rebars are designed separately, and the bar numbers of them sometimes will have conflicts when stirrups with more than two legs are used. For continuous beam that is more common in real-life projects with more than one span, usually they will be considered together during design process, which means the number of sections taken into consideration will rise in proportion as the number of spans increases. In a word, treating the rebar detailing design in each section as a variable, the rebar detailing design is a high-dimensional problem with design variables highly correlated to each other.

2.2. Automated design optimization for steel reinforcement

Computer-aided rebar design has been explored since last century due to the imperative need of the construction industry to efficient and

accurate rebar design [3], estimation [18] and prefabrication [19]. Some research efforts have been made to automatically optimize rebar design using metaheuristic algorithms (MA). However, MAs are generally time-consuming and cannot satisfy the practical use. Neural networks were found capable of reducing time required for design process [1]. This is because neural networks can be trained using a large number of optimal designs, and when a design problem is given, they will come up with a solution instead of going through the design process.

2.2.1. Metaheuristic algorithms

In previous studies on rebar design optimization, MAs are commonly used for different types of RC elements including beams [3,15], columns [20], slabs [2] and shear walls [5]. Most of these algorithms are population-based optimization algorithms like GAs with mechanism inspired by biological evolution, such as selection, crossover and mutation. They have proven effective and robust in solving NP-hard problems with their parallel mechanism and good expansibility to combine with other algorithms [21,22]. However, the disadvantages of applying MAs to rebar detailing design optimization are also obvious. These algorithms usually start with a randomly generated initial

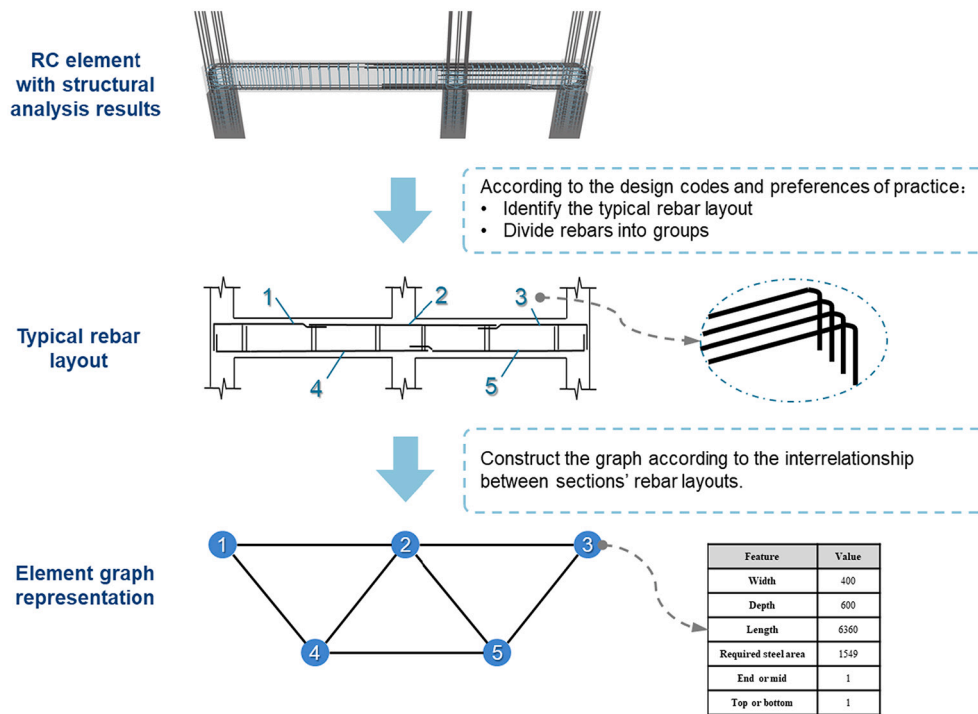


Fig. 3. Graph construction I: preliminary rebar layout analysis – example of two-span continuous beam.

population, which means that previous design experience cannot be inherited for new design cases. Besides, most existing MA-based rebar design optimization approaches are developed at element level [2,5,20], and with the non-negligible computation time required, the overall time consumption may be unbearable in real-life projects with thousands of elements.

2.2.2. Graph neural networks

Graph Neural Networks (GNNs) have received a lot of attention in recent years for their ability to analyze graph-structured data. They are essentially a class of neural networks, but the biggest difference from traditional neural networks is that they can be used to process data represented in graph structure (which is also called graph). Graph, which consists of nodes and edges between nodes, has the potential to denote a large number of systems across different domains, such as social networks [23], physical systems [24], traffic networks [25], and knowledge graphs [26]. The rediscovery of GNNs is attributed to recent advances in deep neural networks, especially convolutional neural networks (CNNs) [12,13]. CNNs are capable of extracting multi-scale spatial features and learning highly expressive representations, which led to the breakthrough in almost all machine learning fields [27–29]. As the generalization of CNNs on graphs, GNNs can operate on not only regular Euclidean data like images (2D grids) and texts (1D sequences) required by CNNs, but also non-Euclidean data like social networks and molecular structure. Therefore, GNNs are adopted in this paper to process the graphs representing rebar layouts, which are a kind of non-Euclidean data with variable sizes.

At present, there are very few applications of GNN in the field of architecture, engineering, and construction. Nauata et al. [30] encoded the constraints of the house layout problem into the graph structure of its relational networks, and applied GNN to aggregate the information of adjacent nodes in the rational networks and construct the feature vector to generate various of floorplans with high quality. Chang and Cheng [31] formulated the building structures as graphs by simplifying the components as joints and their connectivity as edges, based on which GNNs were trained to understand the structural layout, predict structural simulation results, and propose the optimal cross-sections of

elements. Results found that the GNN-based method could dramatically reduce the optimization time required by GA. Gan [32] applied GNN to process the graph data model for volumetric modules with spatial attributes, topological relationship, geometrics and semantics. The information extracted by GNN is then utilized to support automatic generative design of modular buildings. Though GNNs have shown their potential in solving design problems in the field of architecture and civil engineering, the application of GNN in the steel reinforcement design is still lacking.

2.3. Clash diagnosis and resolution

Rebar detailing design in current practice does not involve determining the position of each rebar and considering the intersection between elements, which means that the rebar clashes cannot be identified in the design stage [9]. Rebar clashes are likely to occur at the most congested locations, such as beam-column joints with rebars extending from different elements connecting to the joint, and these clashes are generally identified and resolved manually by workers on site, which is tedious and time-consuming. Moreover, not all clashes can be resolved for a given rebar layout without revising the rebar detailing design [3].

In most systems like architectural, structural and MEP systems, conducting proactive clash avoidance in design phase is widely regarded as more effective and less time-consuming than identifying and resolving clashes passively in construction phase. Existing research [33,34] and even some commercial software like Naviswork [35] can already identify clashes between rebars, though automatic clash resolution approach that is practical and recognized by the industry is still lacking. Hu et al. [36] provided an automatic approach to identify optimal clash correction sequence for MEP system, which could reduce rework and facilitate design coordination. Automatic clash resolution is realized for MEP system by modifying the geometry [37] or location [38] of one of the two clashing components, but the methods are not suitable for rebar clashes because modification of rebar geometry or location may lead to non-compliance with design standards. Liu et al. [39] automatized the clash resolution by employing multi-agent reinforcement learning and BIM. However, the method could not guarantee

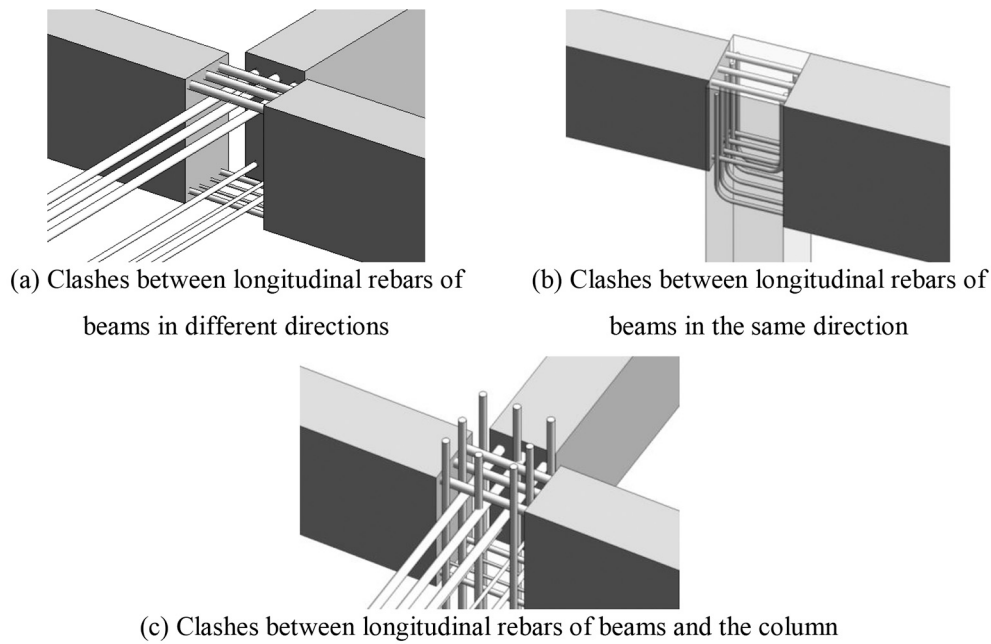


Fig. 4. Three main types of clashes in RC beam-column joints.

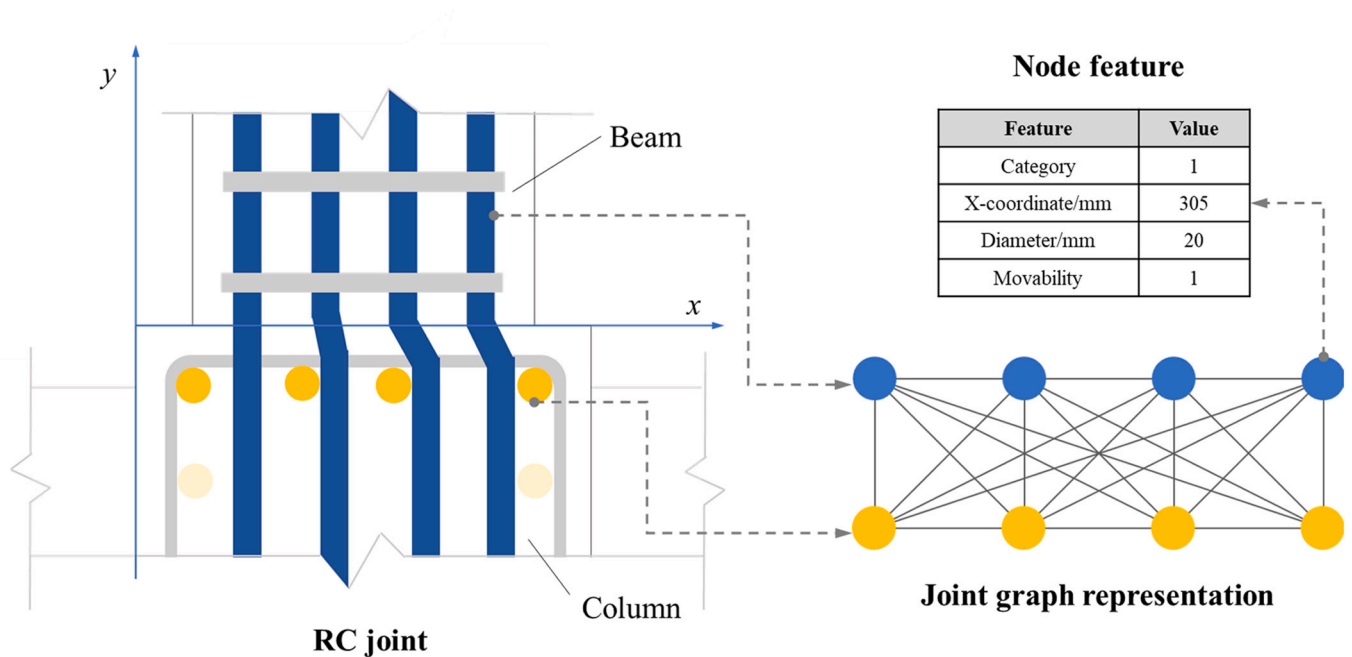


Fig. 5. Graph construction II: joint analysis.

that the rebar can satisfy the standard shape code of the industry, which diminishes the applicability.

3. Methodology

The proposed framework is shown in Fig. 2. The framework involves two GNNs: one for rebar design proposal (RDP-GNN), and one for clash diagnosis (CD-GNN). When training the two GNNs, DfMA principles are also included to enhance the buildability of rebar designs. Starting from the BIM model, the structure model is achieved by idealization and used for structural analysis. With preliminary rebar layout analysis, which is to identify the typical rebar layout of each element as well as the

interrelationship between sections' rebar layouts, the element's rebar layout is represented as a graph (element graph representation), and structure analysis results are then embedded into these graphs. These graph representations are then passed to the RDP-GNN, which will provide initial rebar designs according to the graph structure and also the information embedded. At the same time, the beam-column joints in the BIM model are analyzed and represented as graphs (joint graph representation). The initial rebar designs of elements are then integrated into the joint graph representations. Design checking and optimization are then carried out based on the initial rebar designs using a meta-heuristic algorithm. During the process, rebar designs in the neighbor of the initial rebar design are evaluated for each element and checked to

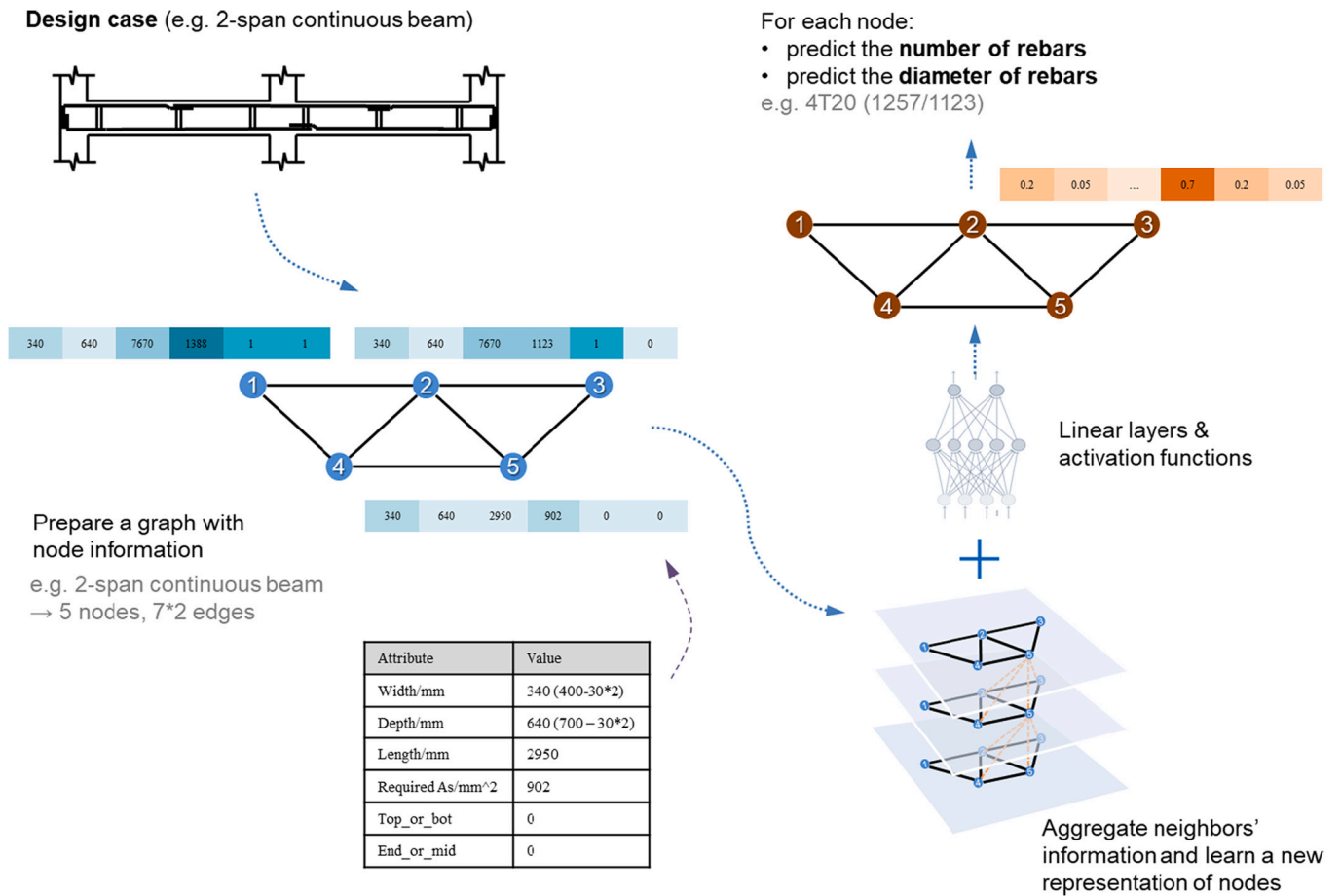


Fig. 6. Workflow of GNN-based rebar design – an example of a 2-span continuous beam.

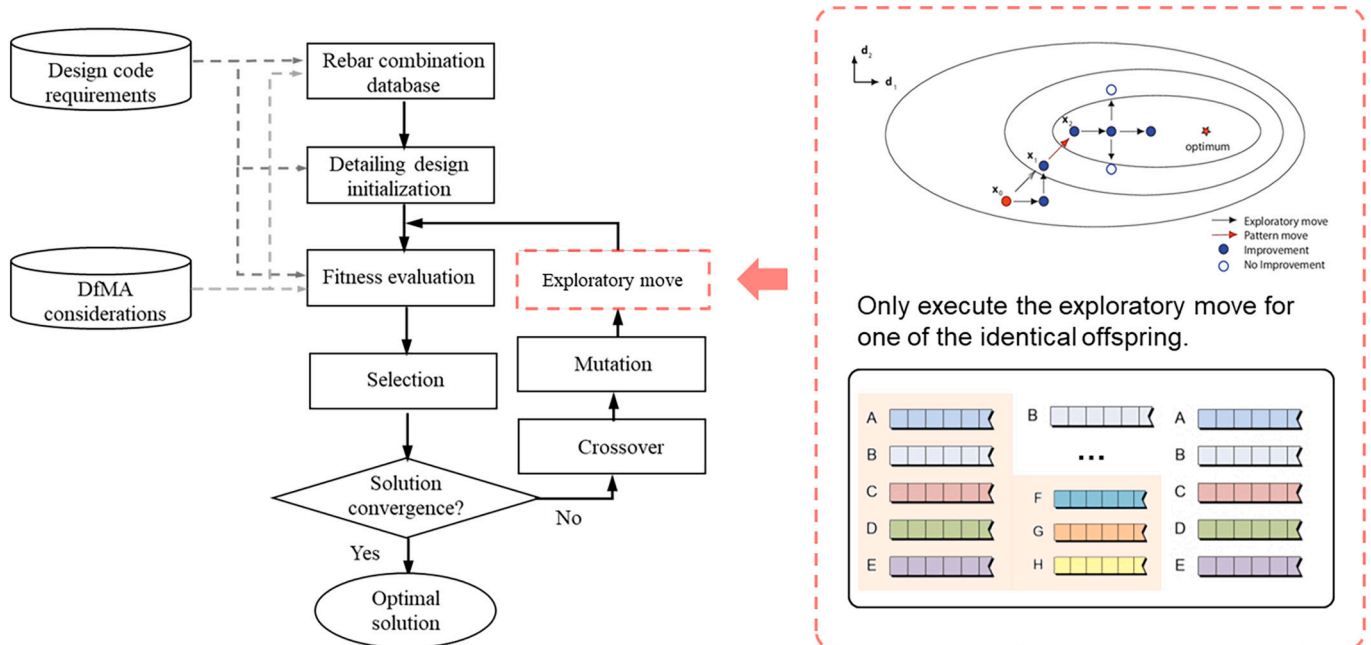
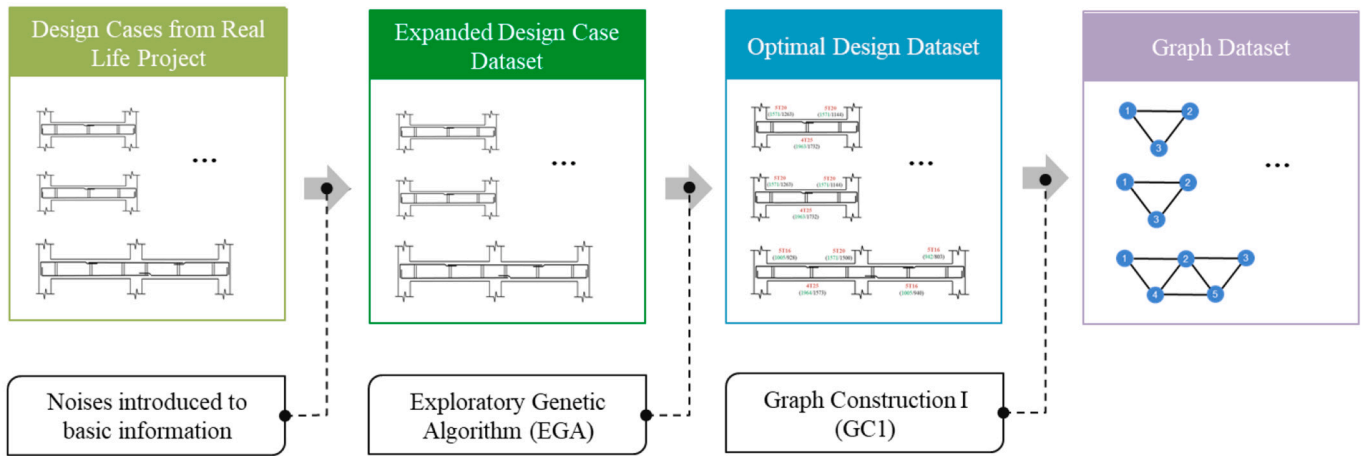
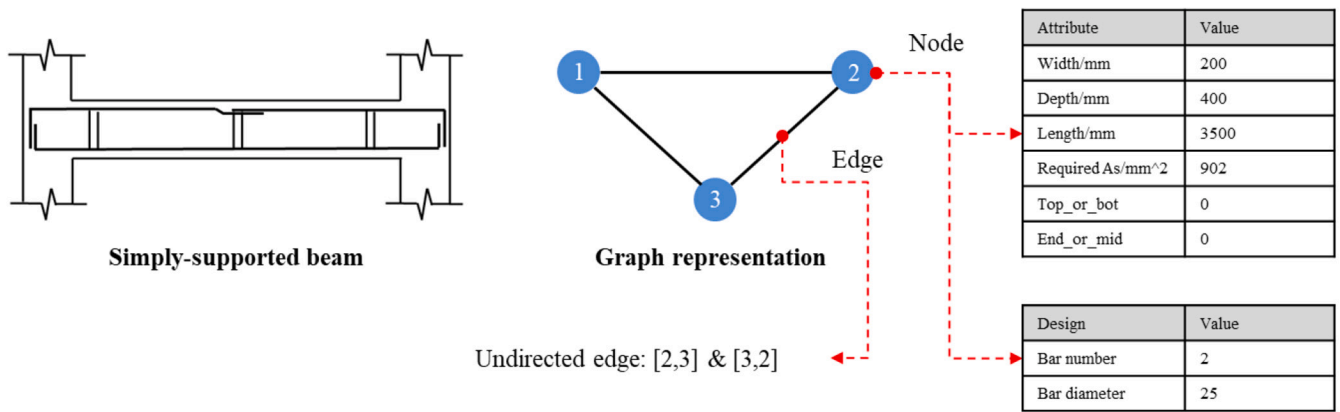


Fig. 7. Method for GNN training dataset preparation.



(a) Dataset preparation process



(b) Example of graphs in the dataset

Fig. 8. Dataset preparation for rebar design proposal graph neural network (RDP-GNN).

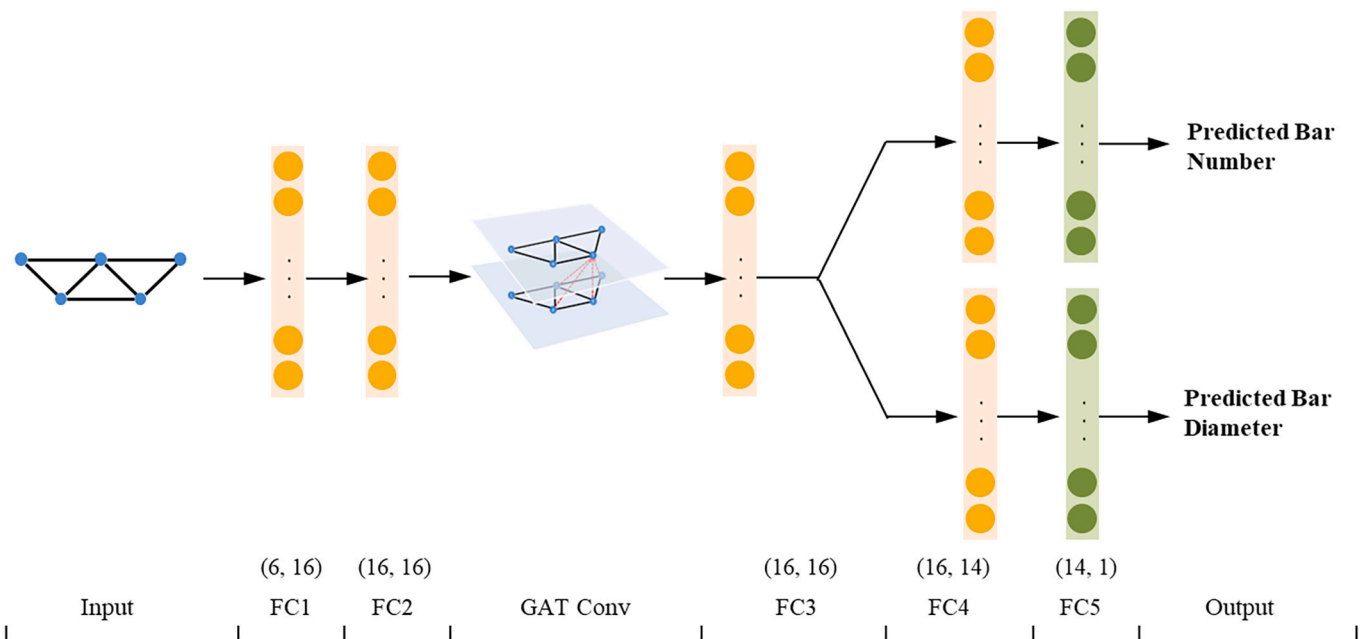


Fig. 9. RDP-GNN model architecture.

Note: 1) 'FC' denotes fully-connected layer, and the numbers in the brackets refer to the number of inputs and outputs, respectively. 2) The activation function used in the GNN for all layers (except FC5) is ReLU function [42], which is not shown in the above figure.

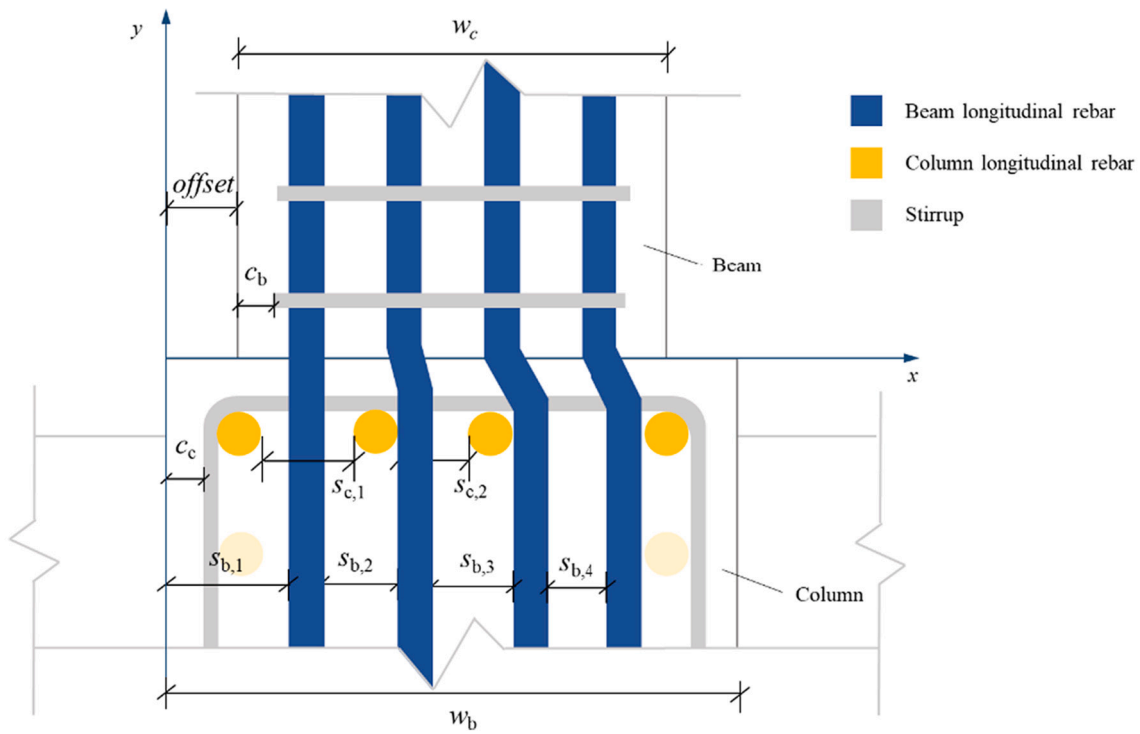


Fig. 10. Formulation of the clash diagnosis problem.

satisfy all the code-stipulated requirements, and those code-compliant designs are retained and sorted according to their total costs, which constitute a feasible rebar design set. The optimal rebar design is set as the default design for each element, and clash diagnosis is conducted for each joint using the CD-GNN. When clashes are identified, rebar design in some elements will be replaced by the design in the design set to solve the clash. The process may take several iterations to ensure a clash-free optimal rebar design, which is then used to update the BIM model. Details of key modules in the framework will be introduced in Sections 3.1–3.4.

3.1. Graph representations of rebar

To develop the GNNs, steel rebars have to be formulated and represented properly by graph representations. In this study, the graph representations consist of two salient components: (1) the first graph represents the rebar layout embedding corresponding design information to support the global exploration of optimal layout design, and (2) the second graph represents RC joints for analyzing the connectivity of steel reinforcement in clash diagnosis.

3.1.1. Graph representation of rebar layout

Graph construction I (GC1) module is to automatically transfer a given RC element into graph representation integrating the design information and the typical rebar layout for GNN-based rebar design proposal. As shown in Fig. 3, this module first identifies the typical rebar layout and divides rebars into groups, according to the design codes and preference of practice. For example, for a two-span continuous beam shown in, its typical rebar layout can be considered to consist of 5 rebar groups, which can be treated as nodes in graph presentation. Then the design-related information of each rebar group like geometrical, positional information, etc., is integrated into the graph. In this paper, each node has its 6 features, including width, depth, length, required steel area, “end or mid” and “top or bottom”. “End or mid” indicates the position of rebars from the horizontal perspective, while “top or bottom” indicates the position from the vertical perspective. These two features

are dummy variables that can only take value 1 or 0. For “end or mid”, “1” implies that the rebar group needs to be anchored into the end supports (like Group 1, 3, 4, and 5), while “0” implies that the rebar group lies between the end supports and no anchorage is needed (like Group 2). For “top or bottom”, “1” refers to top rebars (like Group 1, 2, and 3), while “0” refers to bottom rebars (like Group 4 and 5). The element graph representation can be constructed by linking the nodes with edges according to the interrelationship between different groups of rebars as discussed in Section 1. The edge between a pair of nodes implies that the rebar design of these two groups of rebars will affect each other.

3.1.2. Graph representation of RC joint

The main function of graph construction II (GC2) module is to automatically analyze the connection relationship of rebars at building joints in the BIM model, and get the corresponding graph representation for clash diagnosis. Clash diagnosis is to judge whether a RC joint with given rebar layouts have clashes, and whether these clashes can be solved if yes. This paper first divides rebar clashes into two classes: solvable and unsolvable clashes. Since the space at beam-column joint is very limited, and rebars need to satisfy minimum spacing requirements, sometimes rebar clashes cannot be resolved with code compliance even by bending the rebars, which are defined as unsolvable clashes in this paper. In general, clashes in RC beam-column joints can be classified into following three types: (1) clashes between longitudinal rebars of beams in different directions, (2) clashes between longitudinal rebars of beams in the same direction, and (3) clashes between longitudinal rebars of beams and the column, as shown in Fig. 4. The first type of clashes can be easily resolved by slightly moving the longitudinal rebars of one beam upward or downward, which implies that this kind of clashes is always solvable. In this regard, when judging whether clashes at a beam-column joint could be solved or not, it is reasonable to focus on the clashes between beam rebars and column rebars rather than the clashes between rebars from different beams.

Fig. 5 provides an example of a typical beam-column joint to illustrate how GC2 transfers the rebar layouts at joints into revised bipartite

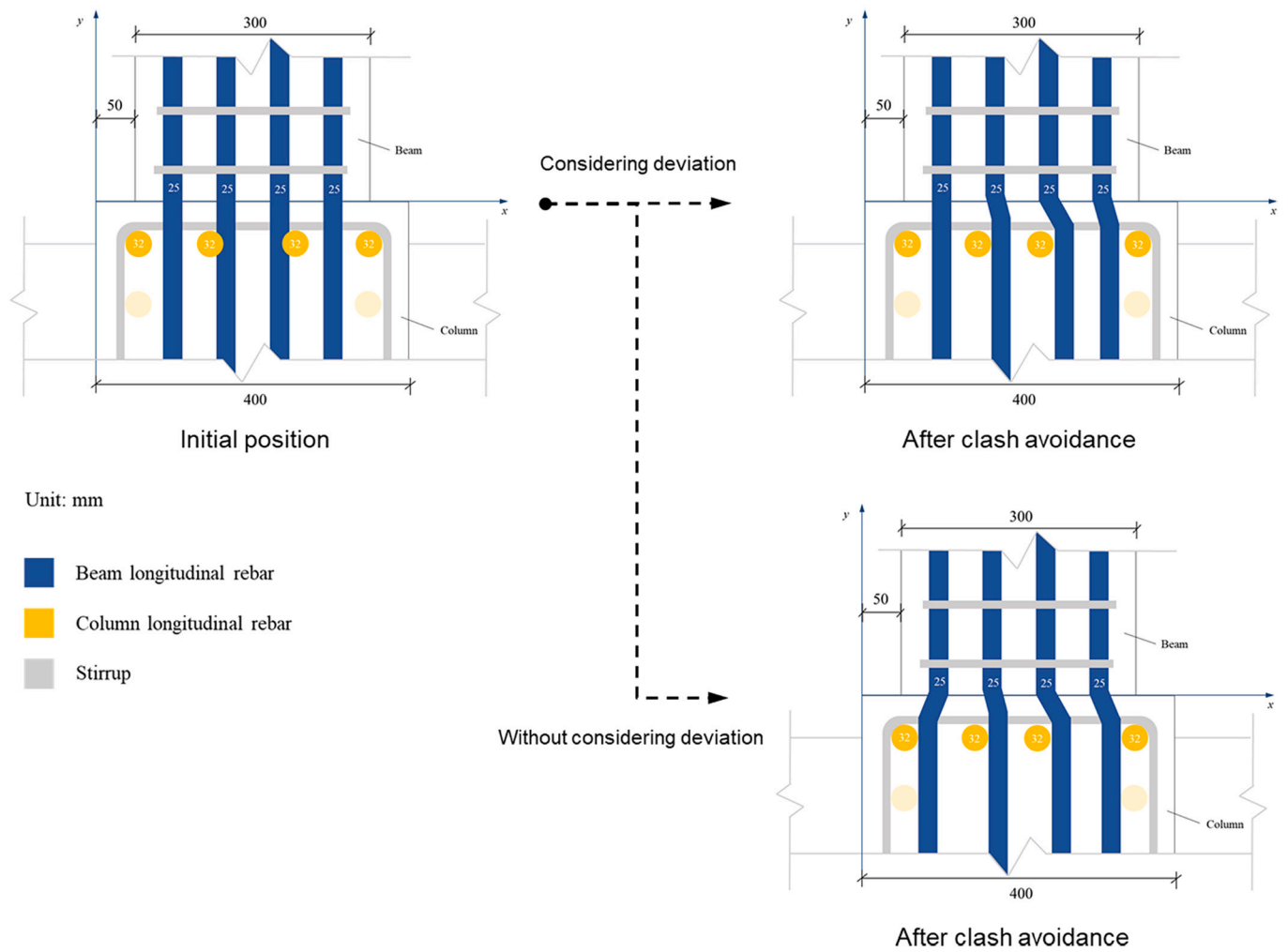


Fig. 11. Advantage of considering the deviation from the initial positions of rebars.

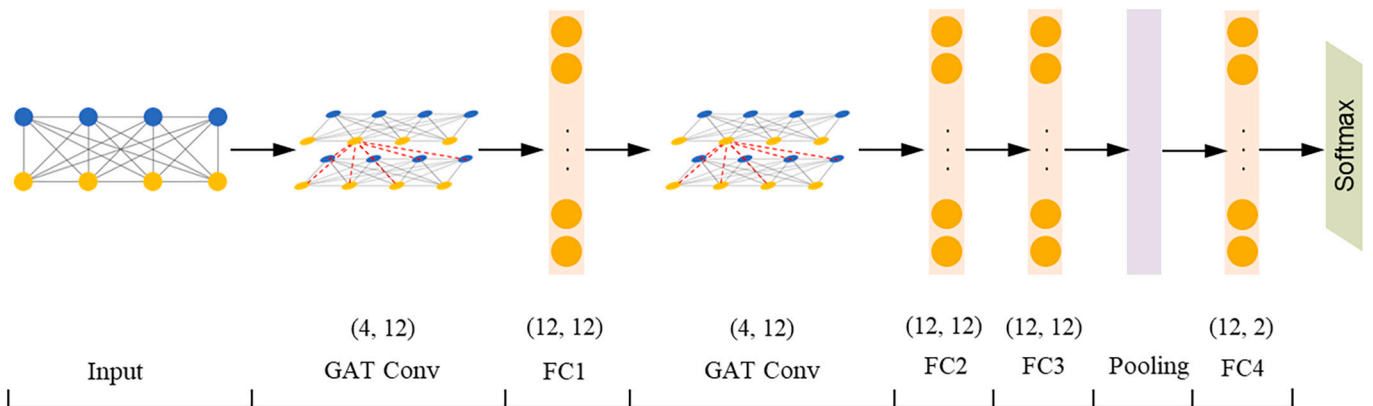


Fig. 12. CD-GNN model architecture.

Note: 1) 'FC' denotes fully-connected layer, and the numbers in the brackets refer to the number of inputs and outputs, respectively. 2) The activation function used for GAT layers, FC1, FC2 and FC3 is tanh function, while the one for FC4 is sigmoid function.

graphs. In the graph representation, each node represents the position of a rebar, while each edge represents the relationship between two rebars. Each node has 4 features including category, x-coordinate of rebar center, rebar diameter, and movability. The "category" is a dummy variable defining which type of RC elements the rebar belongs to. To be specific, the value "1" for "category" means that the rebar is a beam

longitudinal rebar, while "0" refers to a column longitudinal rebar. As shown in Fig. 5, the positions of rebars in the y direction will not affect the existence of clashes at joints, therefore the y-coordinate is not considered as a node feature. "Movability" defines whether a rebar can be moved during the clash diagnosis, which is a reflection of the construction practices since construction workers tend to slightly change

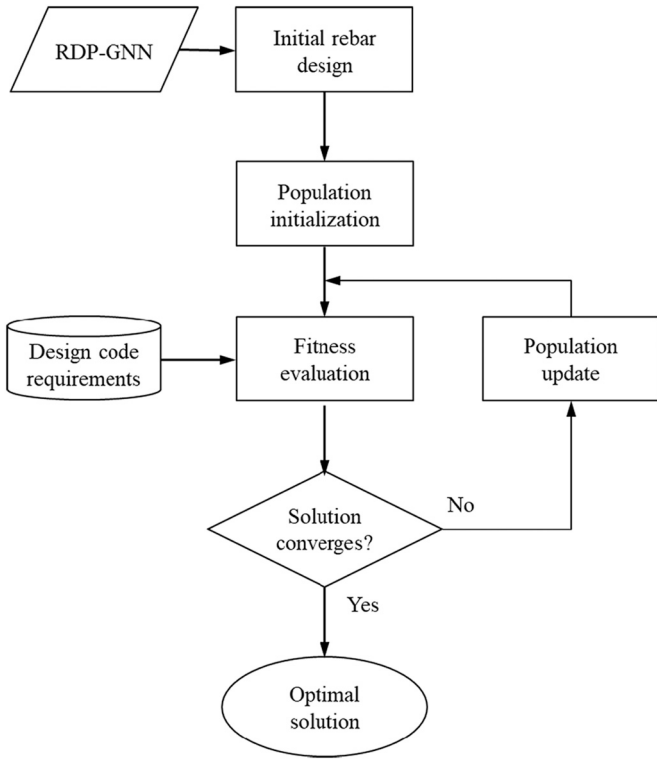


Fig. 13. Hybrid mechanism combining GNN and GA for rebar design checking and optimization.

the positions of rebars to avoid clashes. In this process, the outer two longitudinal rebars of columns should be kept at the corners of stirrups so they are not movable. The edge between two parts of nodes means that the positions of column rebars should not overlap with that of beam rebars, while the edges among one part of the graph indicate the code-stipulated spacing requirements between the rebars should be satisfied.

For multi-layer rebar layouts, the proposed joint graph representation is also applicable since the inner layer of rebars will always align with the outmost layer, and rebar diameter of the inner layer of rebars will not exceed that of the outmost layer, which means that the outmost layer is usually the most congested part and solving clashes in this layer can ensure no clash in the whole section.

3.2. Graph neural network for rebar design proposal (RDP-GNN)

Provided the graph representation of steel reinforcement, a GNN is applied to learn from previous rebar design examples in order to rapidly propose an initial rebar layout for a new design case. To start with, a dataset with thousands of rebar design cases is established for the training of the GNN. Following this, the GNN model is trained properly, during which the network architecture and hyperparameters are fine-tuned. A well-trained GNN can efficiently identify the possible regions where optimal rebar design can be found with a high degree of confidence and thereby accelerate the global exploration process. The initial design is then passed to clash diagnosis by another GNN, and further checking and optimization by EGA to achieve the optimal design.

With the pre-trained RDP-GNN, the initial rebar design of a RC element is proposed in the following steps, as shown in Fig. 6 (taking a two-span continuous beam as an example). First, given a design case, a graph with node feature information is constructed accordingly, where the features of each node are as described in Section 3.1. Then the graph is passed to the pre-trained GNN with graph convolutional layers, linear layers, and activation functions to aggregate each node's neighbors' information and learn a new representation. Finally, based on the new graph representation, the rebar number and diameter are predicted for

each node. In another word, the rebar layout is predicted for each rebar group.

3.2.1. Exploratory genetic algorithm (EGA)

As mentioned above, the training of the GNN for rebar design proposal needs to be based on a large dataset with abundant rebar design cases together with corresponding optimal designs. This paper employed the DfMA-oriented rebar design optimization approach proposed in [9] to consider both material cost and installation cost for rebar designs. The GA-based optimization algorithm combining Hooke-Jeeves method (GA-HJ) in [9] is also improved in this paper to provide optimal rebar designs with higher efficiency. Hooke-Jeeves method (HJ) [40] is a numerical optimization method to find the best match in a multi-dimensional analysis space of possibilities, which does not require a gradient. It includes two kinds of mechanisms for the optimal solutions searching: exploratory move and pattern move. Exploratory move tests the original point in one dimension only, while pattern move searches in the direction from the original point to the new point (generated by the exploratory move). However, the algorithm is time-consuming, since the exploratory move is conducted on each individual in each generation, and pattern move is barely effective for the rebar design problem. Besides, due to the GA selection operator, sometimes there are some identical offspring in one generation, and it seems meaningless to execute a deterministic algorithm like HJ for every identical offspring. Therefore, improvement is made in this paper for higher efficiency by only executing the exploratory move for one of the identical offspring, which is called the exploratory genetic algorithm (EGA) in this paper, as shown in Fig. 7. Detailed comparison between EGA, GA and GA-HJ is provided in Section 4.2.

3.2.2. Graph dataset preparation

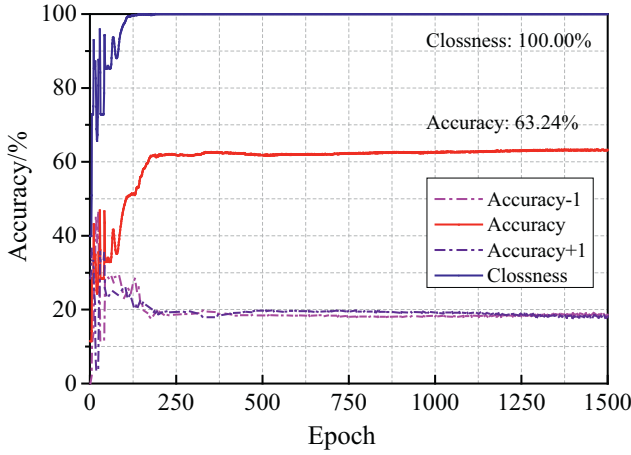
The developed EGA is then used to generate optimal rebar designs to establish the dataset with 2500 design cases, as shown in Fig. 8(a). Hundreds of design cases with elements' basic information including geometry, structural analysis results, etc., were collected from real life projects, and some noises were introduced to these basic cases to generate more design cases by changing the values of those above-mentioned information. Both the basic cases and generated cases were then passed to the EGA to search for the optimal rebar design. For each design case with its corresponding optimal designs, GC1 module represented it as a graph, which is a sample in the data set for the GNN model training. Fig. 8(b) provides an example of the graphs in the dataset. The simply-supported beam is transferred into a graph with nodes integrating the beam's information and edges showing the typical rebar layouts, while the design for each node (bar number and diameter) is also included as the targets.

3.2.3. The RDP-GNN model architecture

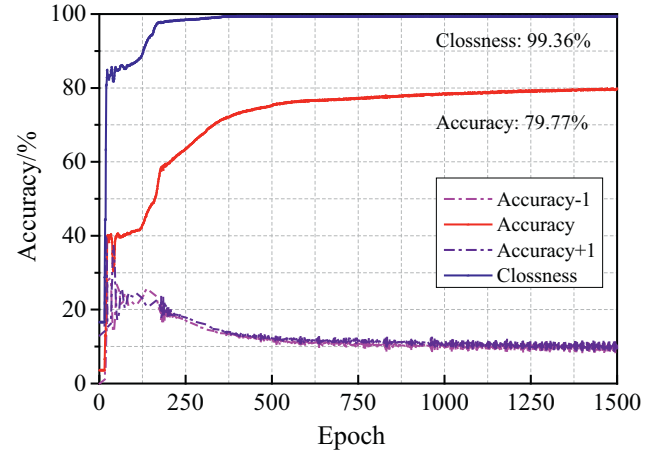
The detailed architecture of the GNN is presented in Fig. 9, which includes one stem and two branches for the predictions of rebar number and diameter, respectively. Graph convolutional layers are leveraged to aggregate the neighbor's information for each node and learn a new representation. Different GNNs are tested and graph attention network (GAT) [41] is adopted in this paper, which is a combination of GNN and attention layers. A GAT can leverage the attention mechanism in GNN to specify different weights to different nodes present in the neighbor, addressing some of the shortcomings of the traditional GNNs.

Huber loss function [43] is employed in the proposed GNN to evaluate the training losses for the two outputs. This loss combines the advantages of both mean absolute error (MAE) and mean square error (MSE), and it is less sensitive to outliers in data [44]. As for the overall loss function, since the GNN has two outputs, the training loss is set as the weighted sum of rebar number prediction loss and diameter prediction loss as below.

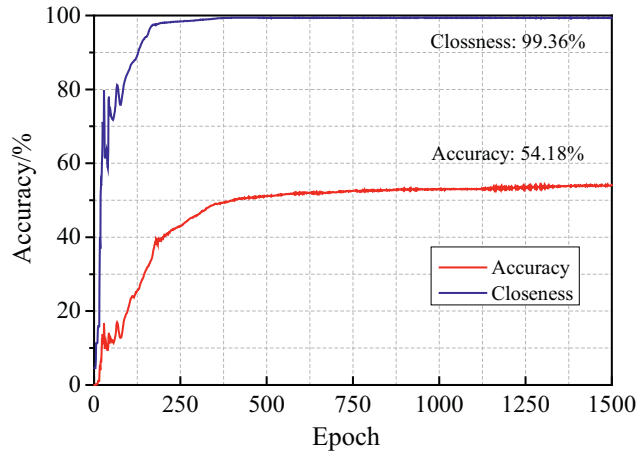
$$L = \alpha \times L_{\text{num}} + (1 - \alpha) \times L_{\text{dia}}$$



(a) Prediction of bar number



(b) Prediction of bar diameter



(c) Prediction of both bar number and diameter

Fig. 14. Training process and performance of the RDP-GNN.

where L , L_{num} and L_{dia} are the overall training loss, the loss for rebar number prediction and the loss for rebar diameter prediction, respectively. α is a weighting factor, which is set as 0.5 in this paper.

The dataset constructed in Section 3.2.2 has 2500 design cases including 1000 simply-supported beams, 1000 two-span continuous beams, and 500 3-span continuous beams. It is split into two parts with a ratio of 80% training set and 20% testing set, and fed to the GNN. The optimizer used is the Adam optimization algorithm [45], and the learning rate is set as 0.002.

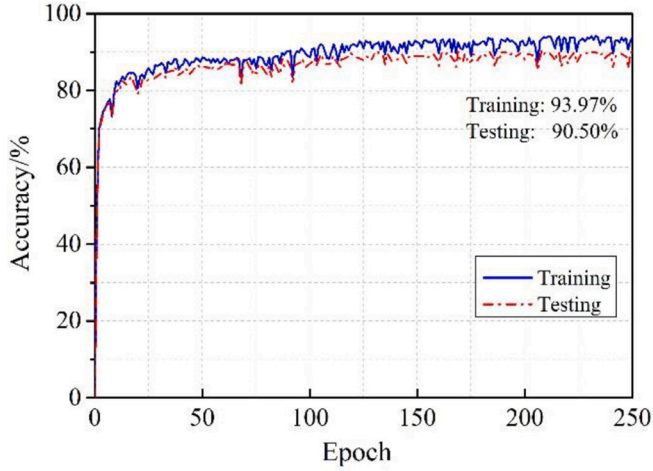
3.3. Graph neural network for clash diagnosis and resolution (CD-GNN)

With the initial rebar design generated by RDP-GNN and the RC joint graph representation, another GNN is trained to conduct clash diagnosis, that is to judge whether there is any unsolvable clash (as defined in Section 3.1.2) in the RC joint with given rebar layouts. Compared to the clash identification and resolution approach proposed in [9] that is time-consuming, the GNN-based method can learn from a large number of examples and identify those unsolvable clashes with ignorable computational time with a pre-trained GNN model, which can significantly enhance the efficiency of rebar design in practices and help to avoid manual clash resolution and rework on construction sites. For the above purpose, the clash diagnosis problem should be properly formulated, and a dataset with abundant cases needs to be prepared for the GNN

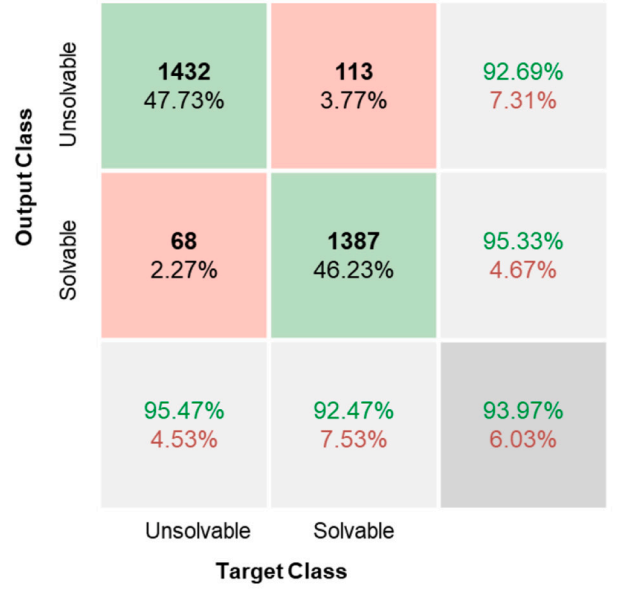
training.

3.3.1. Formulation of the clash diagnosis problem

Following DfMA concept, some assumptions can be made: (1) building workers tend to distribute the rebars evenly in one layer for both beams and columns; (2) it would be more practical if the positions of rebars do not deviate too much from the initial position (by even distribution) after clash resolution; (3) the corner rebars of column cannot be moved. Then the rebar clash diagnosis problem can be formulated as follows. For a given rebar layout, a vector D listing all the diameter of rebars from one side to the other side can be used to describe the layout. D_c and D_b refer to the rebar layout of column and beam, respectively. c_c and c_b refer to the cover of column and beam, respectively. w_c and w_b refer to the width of column and beam, respectively. A coordination system can be established as shown in Fig. 10, where line $x = 0$ is placed at the edge of column. The positions of rebars are variables in the clash avoidance problem. For beam rebars, since each of them can be bent in the joint area, n variables are required to determine a layout with n rebars. For example, three beam rebars in Fig. 10 require 3 variables to specify their positions. As for column rebars, since the corner rebars are fixed, for a layout with $(m + 2)$ rebars, only m variables are required. For example, four column rebars in Fig. 10 require only 2 variables to specify their positions, and the spacing between the third and the fourth rebar will be determined if $s_{c,1}$ and $s_{c,2}$ are determined.



(a) Accuracy in the training process



(b) Confusion matrix

Fig. 15. Training process and performance of the CD-GNN.

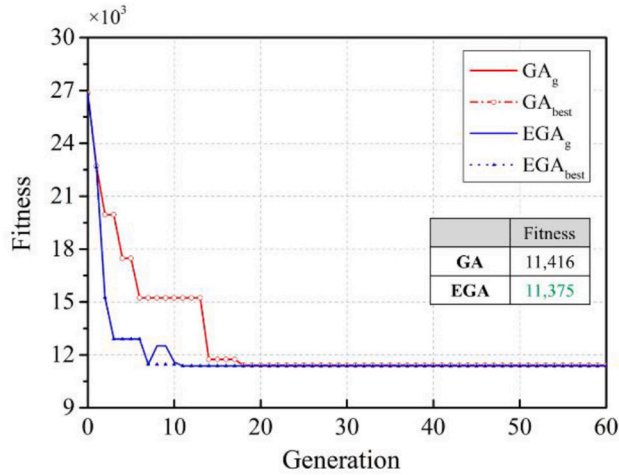


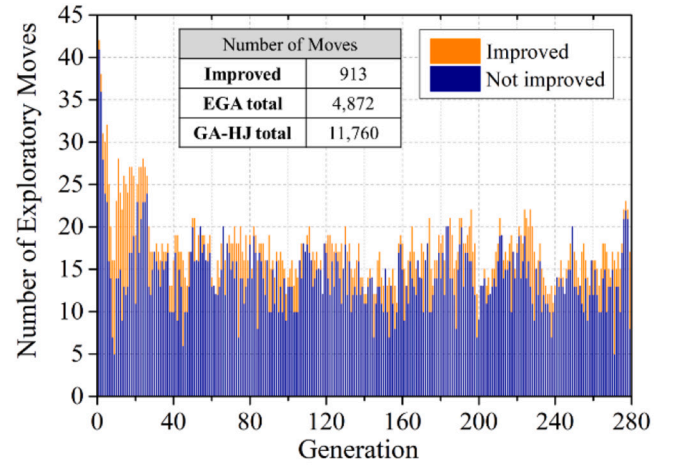
Fig. 16. Improvement of EGA with example of a 3-span continuous beam.

Therefore, the design variables can be expressed as $x = [s_{c,1}, \dots, s_{c,m}, s_{b,1}, s_{b,2}, \dots, s_{b,n}]$. According to the third assumption, a set of initial values of design variables can be achieved by considering even distribution x_0 . A clash index I_c is defined to describe the severity of the clash.

The objective of clash avoidance problem is to check whether this is an unsolvable clash and find the best clash resolution that will not deviate the rebars from their initial positions too much. The objective function and constraints can be expressed as follows.

$$\begin{aligned}
 \min \quad & y = F(x) = (f_1(x), f_2(x)) \\
 & f_1(x) = I_c \\
 & f_2(x) = \sigma(x, x_0) \\
 \text{s.t.} \quad & x(1:m) \geq s_{\min} \quad (\text{bound}) \\
 & \sum_{k=1}^m x(k) + s_{\min} + \sum D_c \leq w_c - 2 * d_{s,c} \quad (\text{constraint}) \\
 & x(m+1) \geq c_c + d_{s,c} + D_c(1) \quad (\text{bound}) \\
 & x(m+2:m+n) \geq s_{\min} \quad (\text{bound}) \\
 & \sum_{k=m+1}^{m+n} x(k) + \sum D_b \leq w_c - c_c - d_{s,c} - D_c(1) \quad (\text{constraint})
 \end{aligned}$$

For given rebar designs, clash-free layout may not be unique, as



shown in Fig. 11. By considering the deviation from the initial positions of rebars, the one that requires the minimum movement of rebars can be identified, reducing on-site rebar bending work and improving the buildability of rebar design, which also comply with the DfMA concept.

3.3.2. Graph dataset preparation

More than one hundred RC joints from real-life projects were investigated and information was collected following the formulation in Section 3.3.1. The design variables in the collected cases were changed within customized and reasonable ranges to expand the dataset. Then the EGA is adopted as optimization algorithm to solve all the cases in the dataset. For each case, the final result will be a dummy value indicating whether unsolvable clash exists. When clash index I_c can decrease to 0 during the evolutionary process, it means no clash exist or all the clashes can be solved, and vice versa. Finally, a dataset with 3000 clash diagnosis cases was established for training the CD-GNN.

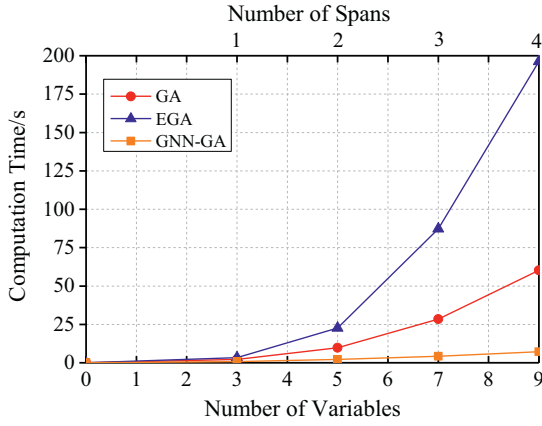


Fig. 17. Efficiency comparison among GA, EGA and GNN-GA.

Note: The algorithm is executed on a laptop with 64-bit system, which includes one CPU of Intel(R) Core(TM) i5-7300HQ @2.50 GHZ and a RAM of 8 GB.

Table 1

Computational time (in second) of GA, EGA and GNN-EGA.

	1-span		2-span		3-span		4-span	
	AVG	STD	AVG	STD	AVG	STD	AVG	STD
GA	2.16	0.25	9.81	0.78	28.43	3.64	60.20	6.41
EGA	3.41	1.06	22.70	5.46	88.12	8.35	195.89	15.44
GNN-EGA	0.78	0.09	2.17	0.47	4.28	2.29	7.23	1.34

Note: AVG refers to average value of computational time, while STD refers to standard deviation.

3.3.3. The CD-GNN model architecture

The CD-GNN is to judge whether a joint with given rebar layouts has unsolvable clashes, which can be regarded as a binary classification task. The model architecture of CD-GNN is shown in Fig. 12. Starting from a joint graph representation, GAT convolutional layers and fully-connected layers with activation functions are applied to aggregate the information for clash classification. Finally, a softmax layer is used to calculate the possibility of two classes – unsolvable and solvable.

The dataset constructed in Section 3.3.2 has 3000 cases including

1500 unsolvable clashes and 1500 solvable clashes. It is split into two parts with a ratio of 80% training set and 20% testing set, and fed to the GNN. The optimizer used is Adam optimization algorithm [45], and the learning rate is set as 0.002.

3.4. Checking and optimization

Since the essence of neural networks is to learn the patterns in the data, GNN-based rebar design prediction cannot guarantee to meet all the code-stipulated requirements such as required steel area and spacing requirements. Therefore, design checking and further optimization are necessary for the initial design proposed by GNN. In this paper, a hybrid mechanism (which is denoted as GNN-EGA) is proposed to optimize the rebar design provided by GNN, as shown in Fig. 13. The initial design is passed to a metaheuristic algorithm for optimization, and the solution space is set as the neighbor of the initial design. Code-stipulated requirements are checked during the process of optimization. Compared to using metaheuristic algorithm along in previous work, the hybrid mechanism makes good use of the experience from previous design cases, which markedly narrows down the solution space and thereby improves the computation efficiency.

4. Experimental results

4.1. Details of training

4.1.1. Training of RDP-GNN

The training process of the RDP-GNN is shown in Fig. 14. Expect for accuracy of prediction, another indicator called “closeness” is defined in this paper, which counts those predictions that are close to the ground truth within a customized value. In this paper, those predictions with a difference less than 1 are counted for rebar number prediction, while those predictions with a difference less than 1 grade are counted for rebar diameter prediction. As shown in Fig. 14 (a) and (b), within 1500 epochs, the accuracy and closeness of bar number prediction reach 63.24% and 100.00%, respectively, while the accuracy and closeness of bar diameter prediction go up to 79.77% and 99.36%. The accuracy of predicting both bar number and diameter is 54.18% while the closeness is 99.36%, as shown in Fig. 14 (c).

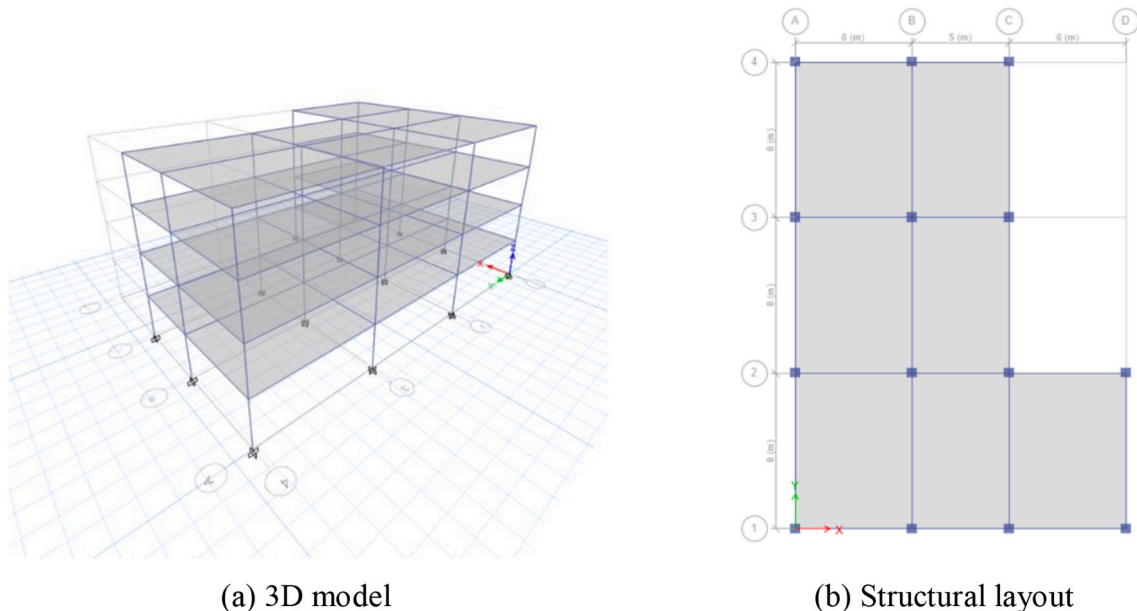


Fig. 18. 4-story RC frame structure subject to uniform floor load and uniform line load.

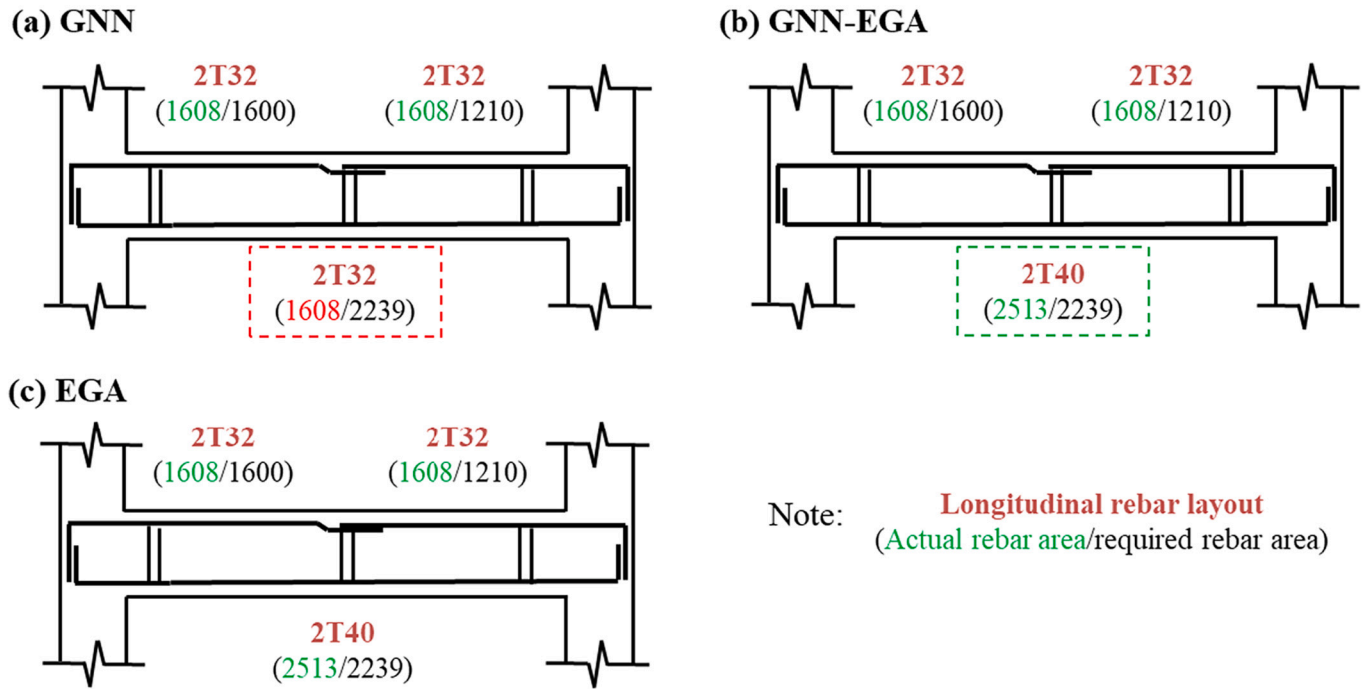


Fig. 19. Comparison of the rebar designs.

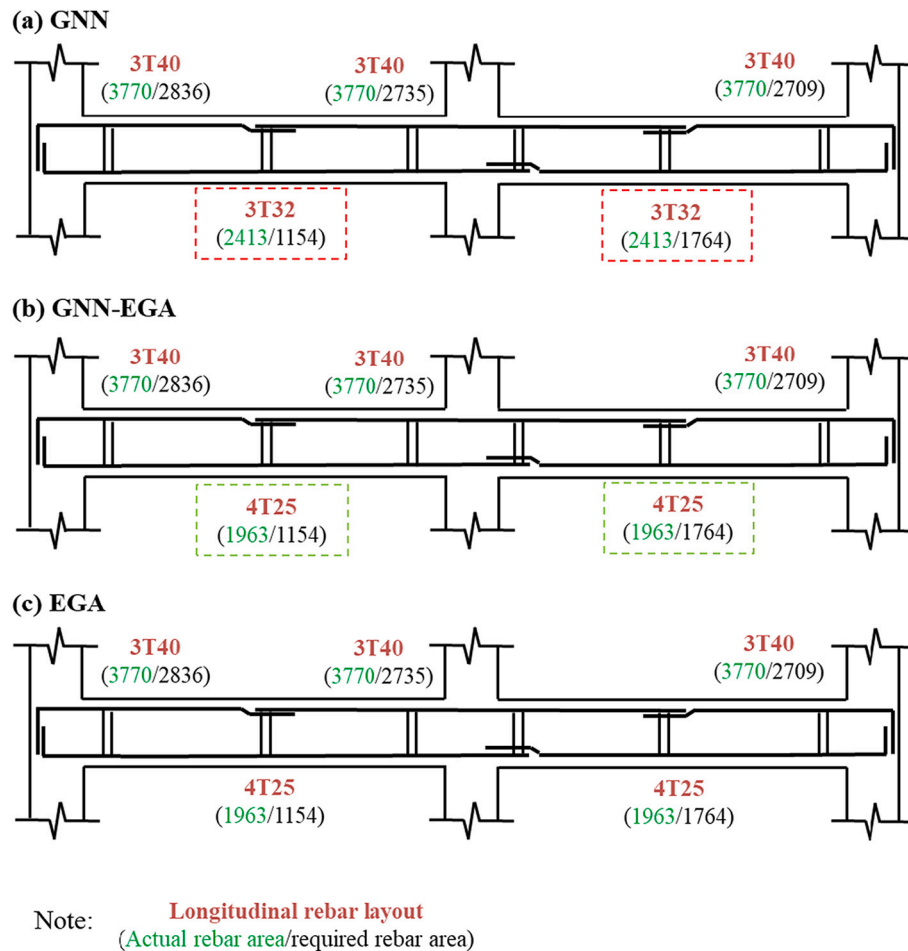


Fig. 20. Comparison of the rebar designs (an example of a two-span continuous beam).

4.1.2. Training of CD-GNN

The training process of the CD-GNN is shown in Fig. 15. As shown in Fig. 15(a), the accuracies of classification go up to more than 90% for both training and testing datasets. The confusion matrix is shown in Fig. 15(b). CD-GNN achieves higher accuracy (95.47%) on unsolvable clashes than solvable clashes (92.47%), meaning that it is more sensitive to those unsolvable clashes and tends to have better performance. This is beneficial to the rebar design in practice, since ignoring an unsolvable clash could do more harm than a solvable clash.

4.2. Evaluation of computational efficiency

The comparison of the EGA, traditional GA, and the GA-HJ developed in [9] is presented in Fig. 16 with an example of a 3-span continuous beam. As shown in Fig. 16(a), compared to traditional GA, EGA can converge to the optimal solution much faster, while GA may be stuck in the local optimum. This is why EGA or GA-HJ is employed for the rebar design optimization rather than GA. Fig. 16(b) shows the number of exploratory moves that improve or not improve the fitness of an individual, and among 4872 exploratory moves, only 913 moves improve the fitness of an individual, which indicates the necessity of cutting down unprofitable moves. In this regard, EGA satisfies the need, which reduces the number of exploratory moves from 11,760 required by GA-HJ to 4872, saving more than 50% of computation time consumed by exploratory moves. In a word, EGA achieves high robustness with relatively less computational time, and is more appropriate for the searching of optimal solution.

However, it is worth noting that the computation time of either GA or EGA is unacceptable for rebar design in real projects with thousands of elements. That is why GNN is employed in this paper to achieve rapid rebar design. Fig. 14 shows that the closeness of both bar number and diameter predictions can approach 100%, which lays the foundation of using metaheuristic algorithm for further design optimization and checking. It means that the optimal solution could be found by searching the neighbor of the design provided by RDP-GNN instead of the whole solution space. In this paper, GA and EGA are applied to optimize the initial rebar design and compared to the GNN-EGA in term of computational efficiency, and the results are shown in Fig. 17 and Table 1.

Fig. 17 and Table 1 show that integrating GNN can save about 75–90% of computational time required by EGA for beams with different spans, and the time saving become more remarkable as the number of variables/spans increases.

4.3. Illustrative example

A two-story RC frame structure (Fig. 18(a)) is used to illustrate the proposed DfMA-oriented approach, and the structural layout is shown in Fig. 18(b). The uniform floor dead load is assumed to be 3.0 kN/m², while uniform floor live load is assumed to be 2.5 kN/m² and 5.0 kN/m² for the first and the second floor, respectively. Besides, uniform line load of 20 kN/m is assigned to all beams. The concrete grades are assumed as C40 for columns and C30 for beams and slabs. The assumed sections for columns and beams are 400 × 400 and 300 × 500 (mm × mm), respectively, while the assumed slab thickness is 150 mm. The rebar unit price is assumed to be 4500 HK\$/ton, while the unit labor cost for rebar installation is assumed as 2500 HK\$/man-day. The available diameters are assumed as 16, 20, 25, 32, 40 for longitudinal rebars and 8, 10, 12 for transverse rebars. Hong Kong COP2013 [46] is used in this example to illustrate the proposed approach, though it can be adopted for building codes in other regions.

Fig. 19 and Fig. 20 provide the rebar designs from GNN, GNN with post-processing algorithm, and EGA. Rebar layout and its steel area are listed compared with the required steel area. The differences in design are highlighted by dashed lines. The results show that GNN alone can generate rebar designs with slight differences with optimal designs, but it cannot guarantee that the design will comply with code requirements.

For example, in Fig. 19(a), the actual steel area of two 32-mm-diameter rebars is only 1608 mm², which is less than the required steel area. Fig. 19(b) and Fig. 20(b) show that by applying metaheuristic algorithm to those designs from GNN for design checking and further optimization, the optimal solution can be achieved.

5. Conclusions and future work

This paper proposes an automatic clash-free rebar design optimization approach with high computational efficiency using GNN and EGA. The framework using GNN for rapid clash-free rebar detailing design is presented, and key modules for graph construction and GNN applications are developed. The graph representations of rebar layouts in both elements and joints are elaborated with examples, which lays down the foundation of future research on rebar using graph-based techniques. Besides, the implementation details are introduced, especially for the GNN-based rebar design proposal and the GNN-based rebar clash diagnosis. An illustrative example is provided, showing that the proposed approach can generate practical rebar layouts with no clash that are comparable to those generated by metaheuristic algorithms. The applications of GNN overcome the drawbacks of using metaheuristic algorithms in the previous study and improves the efficiency of design optimization and clash diagnosis. Moreover, this paper verifies the feasibility of employing deep learning techniques to accumulate previous design experience in the field of rebar design. Sub-optimal rebar design can be provided immediately by a pre-trained network, which markedly narrows down the solution space for post-processing algorithms. A metaheuristic algorithm called EGA is then applied to check and further optimize the design. This hybrid mechanism shows enormous potential for practical use in the industry.

While the results from the illustrative examples are promising and encouraging, the developed approach has three major limitations. Firstly, though the method in this paper is generic and expandable, only one-layer rebar layouts are considered for illustration. Secondly, this paper has not evaluated the clash diagnosis approach for different types of joints except beam-column joints. Last but not least, this paper mainly focuses on RC frame structures with more attentions to the structural design of RC beams and columns. The approaches can be enhanced by incorporating the design optimization of slabs, shear walls, and etc. which would further prompt their applicability for the rebar design of other types of RC structures.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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